Title of the Notes

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November 11, 2026

Just practice problems for Topology, Commutative Algebra, and perhaps Machine Learning and Advanced Algorithms.

1 Exam style stuff

I have to practice for MAT224 Commutative Algebra oral examiniation so these problems are more about intuition, allowing less rigor.

Problem 1.1 (Large algebra identification problem)**.** Let

$$R = k[x, y],\ I = (x^2, xy, y^2),\ A = R/I.$$

What can you say about A? In particular, is it Noetherian or Artinian? Can you give a k-basis or normal form for its elements?
Compute $\sqrt{I}$. Then describe $V(I) \subseteq k^2$. What information about I is visible from $V(I)$?
Is I prime? Is I primary? Give the quickest convincing argument you can.
Determine the maximal ideals of A. Is A local? If so, identify the maximal ideal $\mathfrak{m}$, and compute enough powers of $\mathfrak{m}$ to see what happens.
Compute $\mathrm{Ann}_A(\bar{x})$ and $\mathrm{Ann}_A(\bar{y})$. Are either of these prime ideals?
Consider mutliplication by $\bar{x}$,

$$\mu_{\bar{x}} : A \to A,\ a \mapsto \bar{x}a.$$

Without being given a sequence, identify $\ker \mu_{\bar{x}}$, $\mathrm{im}\, \mu_{\bar{x}}$, $\mathrm{coker}\, \mu_{\bar{x}}$.

Solution. We have $I = (x, y)^2$. We can then say that the ring is Noetherian by using Hilbert's Basis Theorem twice $k \to k[x] \to k[x][y] = k[x, y]$ with Noetherian preserved. Then $I \subseteq R$ is Noetherian so $R/I = A$ is Noetherian.

Artinian does not hold for R since $(x) \supsetneq (x^2) \supsetneq (x^4) \supsetneq \cdots$ never stabilizes. We have an exact sequence

$$0 \to (x, y)(x, y) \to R = k[x, y] \to A \to 0.$$

We have that R is Artinian if and only if $(x, y)^2$ and A are Artinian. We know that R is not Artinian, and $(x, y)^2$ is not Artinian either so we can't conclude anything about A using this.

What we can notice is that any polynomials with x degree ≥ 2 vanish, same for y, while terms in both x and y vanish also. Thus all polynomials have only constant terms, terms in x and terms in y. Thus A is an R-algebra, more specifically a k-vector space with basis $\{1, \bar{x}, \bar{y}\}$.

The root $\sqrt{I}$ is (x, y). We obviously have $(x, y) \subseteq \sqrt{I}$. For $ax + by \in (x, y)$ we have

$$(ax + by)^2 = a^2x^2 + (2ab)xy + b^2y^2$$

which is in I. Now, all generators of I are divisible by by either x or y so $I \subseteq (x, y)$. We have (x, y) prime (and even maximal) so $\sqrt{I} \subseteq (x, y)$ (since $f^n \in I \subseteq J$ implies $f \in J$ due to primality).

We do not have that I is prime, since x^2 is in I, but x is not. It is, however, maximal. We have already established that (x, y) is maximal and $I = (x, y)^2$ so

$$(x, y)^2 \subseteq I \subseteq (x, y).$$

Thus I is (x, y)-primal.

The maximal ideals of A map one to one with the maximal ideals of R containing I.

As mentioned we have $(x, y) \supseteq (x^2, xy, y^2)$ and (x, y) is maximal so

$$(\bar{x}, \bar{y}) = \frac{(x, y)}{(x^2, xy, y^2)}.$$

We can use the third isomorphism theorem to prove that $A/(\bar{x}, \bar{y}) \cong k$.

For uniqueness, suppose $f \in A$. Then $f = a + bx + cy$. The ideal $\mathfrak{m} = (\bar{x}, \bar{y})$ consists of all elements where the constant term is zero, $a = 0$, so an element f not in $(\bar{x}, \bar{y})$ is just $f = a \in k \setminus \{0\}$, which is invertible since k is a field.

As we glimpsed earlier, $\mathfrak{m}^2 = (\bar{x}, \bar{y})(\bar{x}, \bar{y}) = (\bar{x}^2, \bar{x}\bar{y}, \bar{y}^2) = 0$.

We have

$$\text{Ann}_A(\bar{x}) = \{a \in A \mid a\bar{x} = 0.\}$$

Equivalently, in $A = R/I$ we have that the annihilator of $\bar{x}$ are all $a = \alpha + \beta\bar{x} + \delta\bar{y}$ such that

$$a\bar{x} = \alpha\bar{x} + \underbrace{\beta\bar{x}^2}_{0} + \underbrace{\delta\bar{y}\bar{x}}_{0} = 0.$$

We have that $\bar{x}$ is a basiselement of the space, requiring $\alpha = 0$. β, δ can be any elements at all so

$$a = 0 + \beta\bar{x} + \delta\bar{y} \in (\bar{x}, \bar{y}) = \mathfrak{m}.$$

It is not terribly surprising that $\text{Ann}_A(\bar{x})$ is contained in $\mathfrak{m}$, since it is maximal, but actually $\text{Ann}_A(\bar{x}) = \mathfrak{m}$.

We can see this since, for $a\bar{x} + b\bar{y} \in (\bar{x}, \bar{y})$ we have

$$(a\bar{x} + b\bar{y})\bar{x} = a\bar{x}^2 + b\bar{y}^2 = 0.$$

Thus $\mathrm{Ann}_A(\bar{x}) = \mathfrak{m}$, the maximal ideal.

Now, the exact same holds for $\bar{y}$, for the same reasons, and it is not hard to see.

Now for multiplication at $\bar{x}$ we get a natural short exact sequence

$$0 \to A \xrightarrow{\mu_{\bar{x}}} A \xrightarrow{\pi} \mathrm{Ann}_A(\bar{x}) \to 0$$

Problem (Tensor products and quotient modules)**.** Let $R = k[x, y]$, $I = (x^2, xy)$, and $J = (y)$. Put $M = R/I$ and $N = R/J$.
Consider the R-linear map

$$\begin{aligned} \varphi : R &\longrightarrow M \otimes_R N, \\ r &\longmapsto (r + I) \otimes (1 + J). \end{aligned}$$

a) Prove that φ is surjective.
b) Prove that $I + J \subseteq \ker \varphi$.
c) Determine whether the induced map

$$\bar{\varphi} : R/(I + J) \longrightarrow M \otimes_R N$$

is an isomorphism.
d) Determine $M \otimes_R N$ explicitly and exhibit a k-vector-space basis.

Proof of a). Take $\bar{m} \otimes \bar{n} \in M \otimes_R N$. Then there is at least one each of m_0, n_0 such that $\bar{m} = m_0 + I$ and $\bar{n} = n_0 + J$.

Take $m_0 n_0 \in R$. Then

$$m_0 n_0 \mapsto (m_0 n_0) + I \otimes 1 + J = m_0 + I \otimes n_0 + J = \bar{m} \otimes \bar{n}.$$

Proof of b). Let $f \in I + J$. Then

$$f = ax^2 + bxy + cy,\ a, b, c \in k[x, y].$$

Then

$$f \mapsto (ax^2 + bxy + cy) + I \otimes 1 + J.$$

Then

$$(\underbrace{ax^2 + bxy}_{\bar{0}} + cy) + I \otimes 1 + J = cy + I \otimes 1 + J = I \otimes J = 0.$$

Proof of c). Intuitively, φ is surjective with $\ker\varphi = I + J$ (I think) so we know, by the first isomorphism theorem, that

$$R/(I+J) \cong M \otimes_R N.$$

The induced map $\bar{\varphi}$ sends $\bar{r} = r + (I+J)$ to

$$\bar{r} \mapsto \bar{r} + I \otimes 1 + J.$$

As mentioned, $\ker\varphi = I + J$ which corresponds to 0 in $R/(I+J)$. The induced map retains surjectivity as we simply take

$$\bar{m}\bar{n} \mapsto \bar{m} \otimes \bar{n}.$$

■

Solution to d). We have

$$M \otimes_R N \cong R/(I+J) = R/((x^2, xy) + (y)).$$

We have

$$(x^2, xy) + (y) = (x^2, xy, y)$$

so

$$R/(I+J) \cong k[x]/(x^2).$$

We then have a basis $\{1, x\}$ for $k[x]/(x^2)$ which corresponds to the basis

$$\{\bar{1}, \bar{x}\}.$$

■

Extra.

1.

Let

$$\begin{aligned} \beta : (R/I) \times (R/J) &\longrightarrow R/(I+J), \\ (r+I, s+J) &\longmapsto rs + (I+J). \end{aligned}$$

We show that β is well-defined.

Suppose $\bar{r} = r_0 + I = r_1 + I$ with $r_0 \neq r_1$.

Then

$$\begin{aligned}\beta(r_0 + I, s + J) &= (r_0 s) + (I + J) \\ &= \overline{r_0 s} \\ &= \bar{r_0}\bar{s} \\ &= \bar{r_1}\bar{s} \\ &= (r_1 s) + (I + J) \\ &= \beta(r_1 + I, s + J).\end{aligned}$$

The point is that

$$\bar{x}\bar{y} = \overline{xy}.$$

The argument is analogous for $\bar{s}$.

2.

For bilinearity, fix first s, supressing the equivalence class notation for now.

Then

$$\begin{aligned}\beta(r_0(r_1 + r_2), s) &= ((r_0(r_1 + r_2))s) \\ &= r_0(r_0 s + r_1 s) \\ &= r_0(\beta(r_0, s) + \beta(r_1, s)).\end{aligned}$$

We could also be more rigorous than this. The argument for fixed r is the same.

Since β is an R-bilinear function we invoke the tensor products unviersal property, giving a map $\hat{\beta} : R/I \otimes_R R/J \to R/(I + J)$ such that $\beta \circ ((r, s) \mapsto r \otimes s) = \hat{\beta}$. ■

Problem (Compactness and separation axioms)**.** Let X be a compact Hausdorff space.

a) Let $x \in X$, and let $F \subseteq X$ be closed with $x \notin F$. Prove that there exist disjoint open sets $U, V \subseteq X$ such that

$$x \in U, \qquad F \subseteq V.$$

b) Prove that every compact Hausdorff space is normal. That is, if $A, B \subseteq X$ are disjoint closed sets, there exist disjoint open sets $U, V \subseteq X$ with

$$A \subseteq U, \qquad B \subseteq V.$$

c) Let X be an infinite set equipped with the cofinite topology

$$\mathcal{T} = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

Determine, with proofs, whether X is compact, T_1, Hausdorff, and normal.

Proof of a). This follows from compact Hausdorff $\Rightarrow$ normal. $\{x\}$ is closed, since X is Hausdorff, so since X is normal, we can separate

$$\{x\},\ F$$

into disjoint open sets containing them.

Below will be a proof from first principles.

For $y \in F$ we can, by Hausdorffness find

$$U_y \ni x$$

and

$$V_y \ni y$$

disjoint and open.

We have

$$F \subseteq \bigcup_{y \in F} V_y.$$

Since X is compact we have

$$F \subseteq \bigcup_{i=1}^{n} V_{y_i}.$$

Set

$$V = \bigcup_{i=1}^{n} V_{y_i}$$

is open. For x, let U be

$$U = \bigcap_{i=1}^{n} U_{y_i}.$$

This is a finite intersection of open sets so open.

Now

$$U \cap V = \bigcup_{j=1}^{n} \left(\left(\bigcap_{i=1}^{n} U_{y_i} \right) \cap V_{y_j} \right).$$

We have

$$\left(\bigcap_{i=1}^{n} U_{y_i} \right) \cap V_{y_j} \subseteq U_{y_j} \cap V_{y_j}$$

so each term is empty.

Thus $U \cap V = \emptyset$.

■

Proof of (b). We proceed as above. If X is Hausdorff it is T_1.

For fixed $x \in A$ construct

$$V_x = \bigcup_{j=1}^{n} V_{y_i}$$

and

$$U_x = \bigcap_{i=1}^{n} U_{y_i}$$

as in (a) with the property that $U_x \cap V_x$

We do this for all $x \in A$. Then

$$\bigcup_{x \in A} U_x$$

covers A and can be reduced to

$$\bigcup_{i=1}^{n} U_{x_i}.$$

Corresponding V_{x_i}'s independently cover B, so their intersection also does, and it is disjoint to each U_{x_i} and thus disjoint to their union.

■

Proof of (c). It is indeed T_1 since

$$\{x\} = X \setminus (X \setminus \{x\})$$

meaning

$$X \setminus \{x\}$$

is open.

Take two distinct $x, y \in X$. Suppose $x \in U,\ y \in V$ are disjoint open sets.

Then

$$|X \setminus U| = n,\ |X \setminus V| = m$$

for nonzero integers m, n.

Since $U \cap V = \emptyset$ we have

$$U^c \cup V^c = X.$$

Then

$$X \cap (U^c \cup V^c) = X,$$

but then

$$|X| = |X \cap (U^c \cup V^c)| = |X \cap (U^c) \cup X \cap (V^c)| \leq n + m,$$

giving

$$n + m \geq |X| \geq \aleph_0,$$

a clear contradiction.

Thus X is not Hausdorff.

Let $\mathcal{U} = \{U_\alpha\}$ be a cover of some U. Pick some $U_1 \in \mathcal{U}$. Then $X \setminus U_1 = \{x_1, \dots, x_n\}$. For each x_i, if $x_i \in U$, take one of the U_α which contains it. Repeat until all x_i are accounted for. Then we have a cover with at most $n+1$ elements.

The space is not Hausdorff and thus not normal. Suppse it was normal. We know the space is T_1 so $\{x\}, \{y\}$ are closed. Then we can partition them into open disjoint sets, implying Hausdorffness. No good. ■

Problem (Integral extensions)**.** Let k be a field, and consider

$$A = k[t^2, t^3] \subseteq B = k[t].$$

a) State the definition of integrality and determine whether t is integral over A.
b) Show that every element of B has the form $a + bt$, where $a, b \in A$. Deduce that B is a finite A-module.
c) Prove that
$$\text{Frac}(A) = \text{Frac}(B) = k(t).$$
d) Determine whether A is integrally closed in its fraction field.
e) Determine whether every element of B is integral over A, and justify your conclusion.

Proof of (a). Suppose $R \subseteq R'$. $a \in R'$ is integral over R if it is the zero of a monic polynomial

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 x^0, \ a_i \in R.$$

If all elements in R' are monic over R, then R' is integral over R. More precisely, if $R' = R[c_1, c_2, \dots, c_n]$ then R' is an integral extension of R if all c_i are integral over R. The previous is also true since all elements of R are integral over R, take

$$x - a$$

for $a \in R$.

Take

$$f(T) = T^3 - t^3.$$

Then $f(t) = 0$ and

$$f(T) \in k[t^2, t^3][T]$$

so B is integral over A with generating set $\{1, t\}$. ■

Proof of (b). We want to show

$$B = A + At.$$

We have $t^2, t^3 \in A$ so we have

$$t^n \in A$$

for every $n \geq 2$. We could prove this inductively. The base case is trivial. Thereon out,

even powers are easy to see. For $n = 2k$ we have

$$(t^2)^k = t^{2k} = t^n \in A$$

For $n = 2k + 1$ notice that $2k + 1 - 3 = 2(k - 1)$. If $k \geq 4$, then t^{2k-2} exists and thus

$$t^{2k-2}t^3 = t^{2k+1} = t^n \in A$$

exists. So after observing that $t^4 = (t^2)^2 \in A$ this follows.

■

Proof of (c). The forward inclusion $k[t^2, t^3] \subseteq k[t] \subseteq k(t)$ is trivial.

We have that

$$\frac{t^3}{t^2}\frac{t^2}{t^3} = 1,$$

but

$$\frac{t^2}{t^3} = \frac{1}{t}$$

so $(t^3/t^2)^{-1} = t^{-1}$, hence $t^3/t^2 = t$, meaning $k(t) \subseteq \text{Frac}(k[t^2, t^3])$. ■

Problem (Ring, ideal, and module operations)**.** Let $A = k[t^2, t^3] \subseteq B = k[t]$.

a) Determine whether t^5, t^7, t belong to A.
b) Describe explicitly the ideals

$$I = (t^2, t^3)_A, \qquad J = (t^2, t^3)_B.$$

Explain why they are different.
c) Determine A/I and B/J, up to ring isomorphism.
d) Prove that B is finitely generated as an A-module, but that $\{1, t\}$ is not an A-module basis.

Proof of (a). We have that $t^n \in A$ for all $n \geq 2$, so $t^5, t^7 \in A$.

We can prove this inductively by observing that $t^2, t^3, t^4 = (t^2)^2 \in A$.

For even $n = 2k$ it is trivial as

$$(t^2)^k = t^n \in A.$$

By assumption, supposing $n = 2k + 1 - 3 \in A$, which we know since $t^4 \in A$, we can take

$$t^{2k-2}t^3 = t^n \in A.$$

We do not, however, have $t \in A$ as

$$A = k \oplus kt^2 \oplus kt^3 \oplus \cdots$$

Proof of (b/c). The ideal (t^2, t^3) as an ideal of A is the set of

$$at^2 + bt^3,\ a, b \in A.$$

Furthermore,

$$k[t^2, t^3]/(t^2, t^3) \cong k$$

making it maximal.

As an ideal of B we get that

$$B/(t^2, t^3) \cong B/(t^2)$$

where $\bar{t}^2 = 0$ vanishing terms with t-degree greater than or equal to 2.

Explcitly

$$(t^2, t^3) = (t^2).$$

The inclusion $(t^2, t^3) \supseteq (t^2)$ is trivial.

For the other direction, since the ideal absorbs multiplication,

$$t \cdot t^2 = t^3 \in (t^2).$$

So the presence of t kills t^3. When quotienting we see that $\bar{t}^3 = 0$ is redundant since $\bar{t}^2 = 0$ anyways.

Proof of (d). We have t integral over A, and since

$$B = A[t]$$

we have that B is integral over A with generating set $\{1, t\}$.

The generating set is not a basis because it is not linearly independent.

We have

$$t \cdot t^2 - 1 \cdot t^3 = 0$$

so it is not linearly independent over A.

Problem (Product and subspace topologies)**.** Let $X = \mathbb{R} \times \mathbb{R}$ have the product topology, and let

$$P = \{(x, x^2) : x \in \mathbb{R}\}$$

have the subspace topology.

a) State the definition of the product topology and describe its basic open sets.

b) Let

$$U = \{(x, x^2) \in P : -1 < x < 1\}.$$

Prove that U is open in P. Is U open in X?

c) Prove that
$$f : \mathbb{R} \to P, \qquad f(t) = (t, t^2)$$
is continuous, using the universal property of the product topology and the subspace topology.

d) Construct the inverse of f and prove that f is a homeomorphism.

e) Determine whether P is closed in X, and justify your conclusion.

Proof of (a). The product topology, is the coarsest topology for which the universal property holds.

In particular, for continuous f on some space to a cartesian product, f is continuous if and only if
$$\pi_i \circ f$$
is continuous, where π_i denotes the i-th projection.

In the finite case, the basiselements are
$$\prod_{i=1}^{n} U_i,$$
where U_i is open in X_i.

In the infinite case the elements are
$$\prod_{\alpha} U_\alpha$$
where $U_\alpha = X_\alpha$ for all, but finitely many α. In the infinite case this differs from the box topology as the universal property can fail for the box topology. Take for instance $\mathbb{R}^\omega$ with the box topology. Define $f : \mathbb{R} \to \mathbb{R}^\omega$ by
$$t \mapsto (t, t, t, \ldots).$$
For each projection, this is simply the identity map which is continuous. However,
$$\prod_{n=1}^{\infty} (-1/n, 1/n)$$
is open in $\mathbb{R}^\omega$, but the inverse image is just $\{0\}$, which is not open.

■

Proof of (b). We have
$$U = P \cap (-1, 1) \times \mathbb{R}.$$

■

Proof of (c). I do not recall a universal property for the subspace topology.

Some open set in P is of the form

$$P \cap U$$

for open $U \subseteq X$.

Then

$$f^{-1}(P \cap U) = f^{-1}(P) \cap f^{-1}(U) = \mathbb{R} \cap f^{-1}(U) = f^{-1}(U).$$

We have

$$\begin{aligned} f^{-1}(U) &= f^{-1}\left(\bigcup_\alpha V_\alpha \times W\alpha\right) \\ &= \bigcup_\alpha f^{-1}(V_\alpha \times W_\alpha) \\ &= \bigcup_\alpha f_1^{-1}(V_\alpha) \cap f_2^{-1}(W_\alpha), \end{aligned}$$

where $f_i = \pi_i \circ f$.

We have $f_1(t) = t$ and $f_2(t) = t^2$. Thus these are continuous so the preimages are continuous. So we take the union of finite intersections of open sets which is open. This is slightly different from what is asked. We want to take basiselements $\bigcup_\alpha (-\varepsilon_\alpha, \varepsilon_\alpha) \times (-\delta_\alpha, \delta_\alpha)$ and from this clearly see that the inverse image is

$$\bigcup_\alpha (-\varepsilon_\alpha, \varepsilon_\alpha) \cap (-\sqrt{\delta_\alpha}, \sqrt{\delta_\alpha})$$

which clearly is open. ■

Proof of (d). All elements in P are of the form (t, t^2) so

$$f(t, t^2) = t$$

is clearly well-defined.

It is not hard to see that this is continuous either. ■

Proof of (e). The set is closed. Take the continuous $f(x, y) = y - x^2$. We have

$$f^{-1}(\{0\}) = P.$$

■

Problem. Let

$$X = \mathbb{R}^2,\ Q = \{(x, |x|) \mid x \in \mathbb{R}\}$$

with Q carrying the subspace topology.

1. Is

$$W = \{(x, |x|) \mid x > 0\}$$

open in Q? Is it open in X?
2. Is Q closed in X? Prove your answer.
3. Is Q homeomorphic to $\mathbb{R}$? Given an explicit homeomorphism and justify continuity in both directions.

Proof of 1. We begin by checking whether W is open in X.

We observe that, for any $\varepsilon > 0$, the ball $B_\varepsilon(1,1)$ contains

$$q = (1, 1 + \varepsilon/2),$$

but $q \notin W$ so W is not open.

We do however have that W is open in Q.

We have

$$W = Q \cap ((0,\infty) \times \mathbb{R})$$

with the latter being open in X. ■

Proof of (2). Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) = |x| - y.$$

This is obviously a continuous function, and the fiber of 0 is

$$f^{-1}(\{0\}) = \text{all } (x,y), \text{ s.t. } |x| - y = 0.$$

Setting

$$|x| = y$$

it is clear that

$$f^{-1}(\{0\}) = Q.$$

Thus Q is closed since it is the preimage of a closed set under a continuous function. ■

Proof of (c). Define the continuous $f : \mathbb{R} \to Q$ by

$$f(x) = (x, |x|).$$

This map is clearly continuous and well-defined. Now $g : Q \to \mathbb{R}$ by

$$g(x,y) = x$$

is also perfectly well-defined.

Then

$$(g \circ f)(x) = g(x, |x|) = x$$

and

$$(f \circ g)(x, |x|) = f(x) = (x, |x|).$$

■

Problem. Let

$$X = \{0\} \cup \left\{ \frac{1}{n} \,\middle|\, n \in \mathbb{N}_{\geq 1} \right\} \subseteq \mathbb{R}$$

carry the subspace topology induced by the usual metric on $\mathbb{R}$.
Define

$$A = \left\{ \frac{1}{2n} \,\middle|\, n \in \mathbb{N}_{\geq 1} \right\}.$$

a) Show that every singleton $\{1/n\}$ is open in X. Is $\{0\}$ open in X?
b) Determine whether A is open and whether it is closed in X. Justify both.
c) Prove that X is compact, directly from the open-cover definition.
d) Define $h : X \to \mathbb{R}$ by

$$h(0) = 0,\ h(1/n) = (-1)^n.$$

Determine where h is continuous.

Proof of (a). Consider $(1/n - \varepsilon, 1/n + \varepsilon)$, the open ball around $1/n$, in $\mathbb{R}$. This is open for all $\varepsilon > 0$. Choose ε small enough so that $1/(n+1), 1/(n-1) \notin (1/n - \varepsilon, 1/n + \varepsilon)$. Then

$$X \cap (1/n - \varepsilon, 1/n + \varepsilon) = \{1/n\}$$

so $\{1/n\}$ is open due to the subspace topology.

The 0 set however is not open. Suppose $(-\varepsilon, \varepsilon) \cap X = \{0\}$. We can always choose n such that $1/n < \varepsilon$ for any choice of ε, making

$$X \cap (-\varepsilon, \varepsilon) = \{0, 1/n, 1/(n+1), 1(n+2), \ldots\}.$$

This can be made more rigorous, but it should be the right idea. ■

Proof of (b). Once again, take $\varepsilon_n > 0$ such that

$$2(n+1) > 2n + \varepsilon_n > 2n > 2n - \varepsilon_n > 2(n-1).$$

Then

$$X \cap \bigcup_{n \in \mathbb{N}_{\geq 1}} \left(\frac{1}{2n+\varepsilon}, \frac{1}{2n-\varepsilon} \right) = A,$$

so A is open.

It is also closed. We have

$$B := X \cap \bigcup_{n \in \mathbb{N}_{\geq 1}} \left(\frac{1}{n+1}, \frac{1}{n} \right)$$

open and

$$A = X \setminus B,$$

hence A is closed. ∎

Proof. Suppose $\{U_\alpha\} \subseteq 2^X$ is a covering of X. The open sets are

$$U_\alpha = X \cap U$$

where

$$U = \bigcup_\beta U_{\alpha,\beta}$$

where $U_{\alpha,\beta}$ is an interval around some $x \in [0,1]$.

It would be easier to prove that X is a closed subspace of $[0,1]$ which is compact implying X is compact, by observing that

$$X = [0,1] \setminus \bigcup (1/(n+1), 1/n).$$

There must be some open set U_α in the cover containing 0. We can simplify this by thinking of it as

$$X \cap (-\delta, \delta).$$

Then, there exists some $N \in \mathbb{N}_{\geq 1}$ for which

$$1/n \in X \cap (-\delta, \delta)$$

for all $n > N$. For the remaining N $1/n$'s, there must be at least one $U_{\alpha(n)}$ containing it. Thus

$$(X \cap (-\delta, \delta)) \bigcup_{n=1}^{N} U_{\alpha(n)}$$

makes a finite cover. ∎

Proof of (d). We have h continuous at every $1/n$, but $h(1/n) \not\to h(0)$ even when $1/n \to 0$ so there is a discontinuity at 0. ∎

Problem (Localization and quotient modules)**.** Let k be a field and define

$$R = k[x,y]/(xy), \qquad M = R/(\bar{x}^2).$$

Let

$$S = \{1, \bar{y}, \bar{y}^2, \ldots\} \subseteq R.$$

a) Describe M as a k-vector space and determine $S^{-1}M$ up to $S^{-1}R$-module isomorphism.

b) Determine the kernel of

$$\lambda : M \longrightarrow S^{-1}M, \qquad m \longmapsto m/1.$$

Determine whether λ is injective or surjective, with proofs.

c) Let

$$T = \{1, \bar{x}, \bar{x}^2, \ldots\} \subseteq R.$$

Determine $T^{-1}M$, and explain what happens to the relation $\bar{x}^2 = 0$ after localization.

Proof of (a). As we don't have $(xy) \subseteq (x^2)$ we cannot use the third isomorphism theorem.

In M we have that terms in xy vanish and higher terms of x vanish.

For R we can write elements as

$$c + \sum_i a_i \bar{x}^i + \sum_j b_j \bar{y}^j.$$

Quotienting by $\bar{x}^2$ kills higher terms of x giving

$$c + a\bar{x} + \sum_j b_j \bar{y}^j,$$

giving a natural generating set

$$\{1, \bar{x}, \bar{y}, \bar{y}^2, \ldots\}.$$

These are indeed linearly independent since, while $\bar{x}\bar{y} = 0$ seems troubling, we have no problems with sums.

$$a_0 + a_1\bar{x} + b_1\bar{y} + b_2\bar{y}^2 + \cdots = 0$$

has no nontrivial solution.

For $S^{-1}M$ we make all $\{1, \bar{y}, \bar{y}^2, \cdots\}$ invertbile.

For elements of M notice that

$$a_0 + a_1\bar{x} + \underbrace{\sum_j b_j \bar{y}^j}_{\text{each } \bar{y}^j \text{ becomes invertible}}.$$

This leaves the coefficients b_j which themselves are invertible when nonzero.

We identify that

$$\frac{\bar{x}}{1} = \frac{\bar{x}}{1} \cdot \frac{\bar{y}}{1} \cdot \frac{1}{\bar{y}} = \frac{\bar{x}\bar{y}}{\bar{y}} = 0.$$

Thus $\bar{x}$ is killed.

Thus

$$S^{-1}M \cong k[\bar{y}, \bar{y}^{-1}].$$

■

Proof of (b). An element in the localization is 0 if and only if there exists $t \in S$ such that $t \cdot m = 0$ for

$$\frac{m}{s} \in S^{-1}M.$$

We know that $\bar{x}/1 = 0$ so

$$(\bar{x}) \subseteq \ker \lambda.$$

We claim that we also have

$$\ker \lambda \subseteq (\bar{x}).$$

Suppose

$$m \mapsto m/1 = 0.$$

Then there is $a \in S$ such that $a \cdot m = 0$. Elements of M are of the form

$$m = a_0 + a_1\bar{x} + b_1\bar{y} + b_2\bar{y} + \cdots = 0.$$

The only elements in S are powers of $\bar{y}$.

So if

$$\bar{y}^k a_0 + \bar{y}^k a_1 \bar{x} + b_1 \bar{y}^{k+1} + b_2 \bar{y}^{k+2} + \cdots = 0,$$

then the term with $\bar{x}$ vanishes, requiring

$$a_0 \bar{y}^k + \sum_j b_j \bar{y}^{k+1} = 0$$

which we have established have no nontrivial solution in M so m lies in $(\bar{x})$.

Since $S^{-1}M \cong k[y, y^{-1}]$ we cannot have surjectivity. The image is $k[y]$ as there is no element mapping to $1/y$. ■

Proof of (c). We make all $\bar{x}$ invertible so

$$\frac{m}{1} = \frac{1}{\bar{x}^2} \cdot \frac{\bar{x}^2 m}{1} = \frac{0}{\bar{x}^2} = 0$$

so $T^{-1}M = 0$.

■

Problem (Deformation retractions and fundamental groups)**.** Let

$$X = \mathbb{R}^2 \setminus \{(0,0)\}, \qquad A = S^1, \qquad p = (1,0).$$

a) Construct a continuous map

$$H : X \times [0,1] \to X$$

such that

$$H(x,0) = x, \qquad H(x,1) \in A,$$

and $H(a,t) = a$ for all $a \in A$, $t \in [0,1]$. Verify the required properties.

b) Prove that

$$\pi_1(X,p) \cong \mathbb{Z}.$$

You may assume $\pi_1(S^1,p) \cong \mathbb{Z}$. Identify a generator.

c) Identify $\mathbb{R}^2$ with $\mathbb{C}$, and let

$$f : X \to X, \qquad f(z) = z^2.$$

Determine the induced homomorphism $f_* : \pi_1(X,p) \to \pi_1(X,p)$. Determine whether it is injective and surjective.

d) Let $j : X \hookrightarrow \mathbb{R}^2$ be the inclusion. Determine

$$j_* : \pi_1(X,p) \to \pi_1(\mathbb{R}^2,p).$$

Explain why this is compatible with the result of part (b).

Proof of (a). The map

$$H(x,t) = (1-t)x + t\frac{x}{\|x\|}$$

works.

For $t = 0$ we have

$$H(x,0) = x$$

and for $t = 1$ we have

$$H(x,1) = \frac{x}{\|x\|}.$$

For $x \in S^1$, we have $\|x\| = 1$ so this is the identity map here.

Furthermore if $x \in S^1$ then

$$H(x,t) = (1-t)x + t\frac{x}{\|x\|} = x - tx + t\frac{x}{|} = x.$$

For well-definedness notice that $\mathbb{R}^2 \setminus \{(0,0)\}$ is closed under multiplication and addition, $\|x\|$ is nonzero so there are no division by zero issues.

For continuity notice that we are simply doing scalar-multiplication and addition which is continuous. ■

Proof of (b). The existence of H means that S^1 is deformation retract of X, which pre-

serves fundamental group up to usual basepoint shenanigans.

Suppose A is deformation retract of X, meaning $H : X \times I \to X$ exists with the required properties.

Define $r : X \to A$ by $r(x) = H(x, 1)$. Then $r(a) = a$ for all $a \in A$ so

$$r \circ i = \mathrm{id}_A$$

for the inclusion $i : A \to X$.

Then

$$(\mathrm{id}_A)_* = (r \circ i)_* = [r * i] = \mathrm{id}_{\pi_1(A)},$$

but

$$r_* \circ i_* = (r \circ i)_* = \mathrm{id}_{\pi_1(A)}.$$

r_*, i_* are induced homomorphisms

$$r_* : \pi_1(X) \to \pi_1(A),$$

and

$$i_* : \pi_1(A) \to \pi_1(X)$$

whose composition are identity maps. Thus they are mutually inverse isomorphisms letting us conclude that

$$\pi_1(A) \cong \pi_1(X).$$

■

Proof of (c). ■